ESP32 LILLYGO T DISPLAY

Main feautures

Feature	Specification
Processor	ESP32-D0WD-Q6 (Dual-core 32-bit Xtensa® LX6, up to 240 MHz).
Memory	4MB or 16MB Flash (Internal/External SPI) and 520KB SRAM.
Display	1.14" IPS LCD (135×240 px) powered by the ST7789 driver.
Connectivity	2.4 GHz Wi-Fi (802.11 b/g/n) and Bluetooth 4.2 / BLE.
Power	3.3V/5V DC input; includes TP4054 Li-Po charging circuit (500mA).
Peripherals	2x Programmable buttons, USB-C (CH9102 or CP2104 bridge), ADC/SPI/I2C/UART.

Advantages

- **All in one device:** The built-in display removes the need for external wiring/breadboards for UI.
- **Dual-Core Architecture:** Allows for parallel processing (e.g., handling WiFi on Core 0 while running the stress test/GUI on Core 1).
- **Battery Native:** Built-in charging circuitry and a 2-pin JST connector make it ready for portable use.

Disadvantages

- **Thermal Density:** The integrated antenna and CPU are on the same chip; heavy WiFi load generates concentrated heat.

Toolchains & IDEs

- **Arduino IDE:** Standard for beginners; uses the "ESP32 Dev Module" board definition.
- **PlatformIO (CLI/VS Code/Neovim):** Recommended for advanced users. Manages complex libraries like `TFT_eSPI` via the `platformio.ini` file.
- **MicroPython/CircuitPython:** Ideal for rapid prototyping; enables a "USB drive" style workflow similar to the Raspberry Pi Pico.
- **ESP-IDF:** The official C/C++ development framework from Espressif for professional, low-level optimization.

Ease of Getting Started

- **Moderate.** * Unlike the Pico or Nano, the LilyGO requires specific library configurations to see *anything* on the screen. A common failure point for beginners is forgetting to set

GPIO 4 (Backlight) to HIGH. However, once the environment (like PlatformIO) is configured with the correct `build_flags`, development is highly efficient due to the rich community library support.

Ease of Control (Serial and WiFi)

- **Serial Control: Very Easy.** The onboard USB-to-Serial chip (CH9102/CP2104) is recognized natively by Linux. Standard baud rates (115200) work flawlessly for real-time debugging and command input.
- **WiFi Control: Excellent.** The ESP32 is arguably the best-in-class for WiFi at this price point. It supports Station mode (client) and Access Point mode (server) simultaneously. During stress testing, it successfully maintained high throughput (Mbps) while simultaneously updating the display, proving its robustness compared to microcontrollers that use external SPI-based WiFi modules.

display is conencted trough SPI

Pinout

Name	V18
TFT Driver	ST7789
TFT_MISO	N/A
TFT_MOSI	19
TFT_SCLK	18
TFT_CS	5
TFT_DC	16
TFT_RST	23
TFT_BL	4
I2C_SDA	21
I2C_SCL	22
ADC_IN	34
BUTTON1	35
BUTTON2	0
ADC Power	14